

Executive Business Report

End-to-End Sales Forecasting & Demand Intelligence System

1. Executive Summary

This project was developed to help a retail business improve inventory planning by forecasting future sales, identifying unusual sales patterns, and grouping products according to their demand. Historical sales data from four years was analysed using statistical and machine learning techniques. Three forecasting models were evaluated: SARIMA, Prophet, and XGBoost. Based on model performance, XGBoost was selected because it achieved the lowest prediction error. The system also detected unusual sales weeks using anomaly detection and grouped products into demand segments using K-Means clustering. Finally, the results were presented through an interactive Streamlit dashboard so that business users can easily explore forecasts, anomalies, and product demand trends without requiring technical knowledge.

2. Key Findings from Exploratory Data Analysis

The sales data showed an overall upward trend over the four-year period, indicating steady business growth. Seasonal patterns were visible, with higher sales during the year-end festive months. Technology products generated the highest overall revenue, while some regions demonstrated more stable sales growth than others. The average shipping time remained fairly consistent across regions, suggesting an efficient delivery process. These insights provided a strong foundation for building reliable forecasting models.

3. Sales Forecast Results

Three forecasting models were compared using MAE, RMSE, and MAPE. XGBoost delivered the best performance with a MAE of 8,620.99, RMSE of 11,043.31, and MAPE of 12.41%. The expected sales for the next three months are:

- Month 1: 84,931.44
- Month 2: 91,793.08
- Month 3: 88,792.56

The forecast suggests that demand will remain strong over the coming months, with a noticeable increase during the second forecast month before stabilising.

4. Top Anomalies Detected

Three important anomalies were identified:

- 22 March 2015 – Sales increased sharply, likely due to a promotional campaign or bulk orders.
- 17 December 2017 – High sales during the year-end shopping season.
- 2 December 2018 – Significant sales spike, likely driven by festive demand and special offers.

Identifying these events helps the business separate exceptional sales from normal demand patterns.

5. Product Demand Segmentation

Products were grouped into three demand segments:

- High Volume, Stable Demand
- Growing Demand
- Low Volume, Stable Demand

Products with consistently high demand should be prioritised for inventory, while low-volume products should be stocked carefully to minimise holding costs.

6. Business Recommendations

1. Increase inventory for high-demand and growing-demand products.
2. Plan inventory and marketing activities around seasonal demand peaks.
3. Continuously monitor anomalies to identify unusual business events and respond quickly.

7. Limitation of the System

The model relies on historical sales data. Unexpected events such as economic changes, supply chain disruptions, or sudden shifts in customer behaviour may reduce forecast accuracy. The model should be retrained regularly using updated data.

8. Conclusion

The Sales Forecasting and Demand Intelligence System supports better inventory planning by combining forecasting, anomaly detection, product segmentation, and an interactive dashboard. It helps managers make informed decisions, reduce inventory risks, and improve operational efficiency.